MECH 325
Homework Assignment #4

Problem 1 (Mott 16-4 Modified)

a) A bushing of 0.50" diameter is subject to a radial load of 75 lbf and rotates at 600 rev/min. Use an L/d ratio for the bearing in the range of 0.50 to 1.50. Compute a *PV* value and specify a material from Table 16-1. Include the design factor, $n_d = 2$.

b) Compare this selection with the method presented in Shigley. Comment on the result.
The following factors are known or can be assumed:

- $f_s = 0.04$ $\quad\quad$ $\hbar_{CR} = 2.7$ Btu/(hr·ft^2·F)
- J = 778 ft ·lbf/Btu $\quad\quad$ Ambient temperature = 70 F

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a)

$$F = 75 \text{ lb}; D = 0.50 \text{ in}; n = 600 \text{ rpm}$$

$$\text{Let } L = 1.5D = 1.5(0.50 \text{ in}) = 0.75 \text{ in}$$

$$p = \frac{F}{LD} = 75 \text{ lb}/(0.75 \text{ in})(0.50 \text{ in}) = 200 \text{ lb/in}^2$$

$$V = \frac{\pi Dn}{12} = \pi(0.50 \text{ in})(600 \text{ rpm})/12 = 78.5 \text{ ft/min}$$

$$pV = (200 \text{ psi})(78.5 \text{ fpm}) = 15\,700 \text{ psi} \cdot \text{fpm}$$

$$\text{Required } pV = 2(15700) = \mathbf{31400 \text{ psi} \cdot \text{fpm}}$$

Porous bronze or BU bearing

Note: Since the selection of L/D is variable, we can allow other values other than L = 1.5D. For example, L=D solution is below.

a) GIVEN: $D = 0.5$ in
$F = 75$ lbf
$N = 600$ rpm
$0.5 < L/D < 1.5$
$n_d = 2.0$

Select $L/D = 1 \rightarrow L = 0.5$ in

$$P = \frac{F}{LD} = \frac{75\ lb}{(0.5\ in)^2} = 300\ psi \quad \text{[See pg. 620, Mott]}$$

$$V = \frac{\pi DN}{12} = \frac{\pi (0.5 in)(600 rpm)}{12\ in/ft} = 78.54\ ft/min$$

design $PV = n_d \cdot P \cdot V = 2.0(300\ psi)(78.54\ ft/min) = \boxed{47124\ psi\ ft/min}$

From Table 16-1: Select Porous / oil-impregnated bronze : $PV = 50000$ psi-fpm OK

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b) SHIGLEY
- PV method is identical to above : $PV_{design} = 47124$ psi-fpm
- Also check temperature rise :

[Eq. 12-49] $$T_f \approx T_\infty + \frac{720\, f_s\, F N}{J\, h_{CR}\, L} \quad (\times n_d)$$

Given: $f_s = 0.04$
$J = 778$ ft·lbf/Btu
$h_{CR} = 2.7$ Btu/hr ft^2·F
$T_\infty = 70$ F

[Table 12-8]
Try porous bronze : $T_{max} = 150°F$
$V_{max} = 1500$ fpm (OK) ✓
$PV = 50000$ psi-fpm (OK) ✓

Solve for required length :

$$L = \frac{720\, f_s\, F \cdot n_d \cdot N}{J\, h_{CR}\,(T_f - T_\infty)} = \frac{720(0.04)(75\ lb)(2.0)(600\ rpm)}{778\ \frac{ft\,lbf}{Btu}\left(2.7\ \frac{Btu}{hr\,ft^2 F}\right)(150-70)°F} = 15.424\ in\ !$$

$L_{max} = 1.5D = 0.75$ in

$L \gg 0.75$ NOT ACCEPTABLE

Mott does not account for temperature rise in bushing - Shigley suggests a 0.5 in long bronze bushing will get very hot...

($T_f \approx 2540°F$!)

Select new size and new material

Use $L_{max} = 0.75$ in, solve for T_f required of material

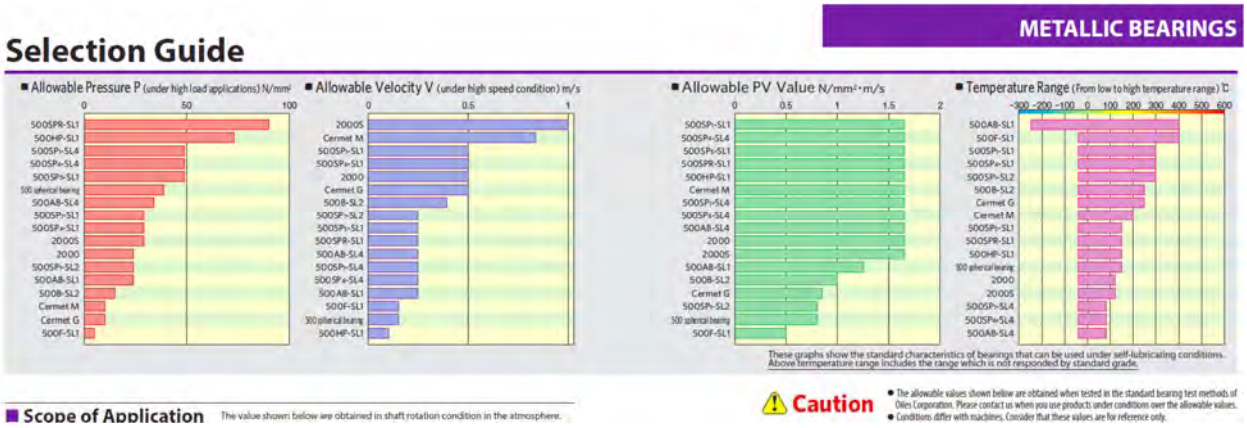
$$T_f = 70F + \frac{720(0.04)(75lb)(2.0)(600\ rpm)}{778(2.7)(0.75\ in)} = 1715°F \text{ required...}$$

No materials from Table 12-8 are suitable for such high temperatures

- must increase L above 1.5 D or increase diameter + length to allow for better heat dissipation

OR - reduce load, friction, operating speed to generate less heat.

Comment: A 600rpm rotational speed with 75 lbs of force on a ½ inch diameter shaft will generate significant heat. It is critical that a heat estimate be performed to establish if a bushing is feasible. From the Oiles catalogue (see Canvas site), no metallic bearings can stand temperatures above 400 °C. In this way the Shigley method is best as you will conclude that either the length of the bushing must increase significantly (not practical) or a roller bearing is better suited for this application (best option). Looking at Table 14-3, a 12 mm. deep grove ball bearing has a minimum dynamic rating of 1140 lbs. It is perfect for this application.



Problem 2 (Shigley Question 12-27)

a)

12–27 Choose an Oiles SP 500 alloy brass bushing to give a maximum wear of 0.002 in for 1000 h of use with a 200 rev/min journal and 100 lbf radial load. Use $h_{CR} = 2.7$ Btu/(h · ft^2 · °F), $T_{max} = 300$°F, $f_i = 0.03$, and a design factor $n_d = 2$. The bearing is to operate in a clean environment at 70°F. Table 12–12 lists the bushing sizes available from the manufacturer.

Assume the shaft diameter is 1.25 inch.

b) Use the Oiles catalogue to confirm the selection in part a) meets the current limits in the catalogue. (See tutorial)

GIVEN: $w = 0.002$ in
$t = 1000$ h $= 60\,000$ min.
$N = 200$ rpm
$F = 100$ lbf
$T_{max} = 300$ F / $T_\infty = 70°$ F
$f_s = 0.03$
$n_d = 2.0$

$D = 1.25$ in
[Table 12-12]:
Available Lengths: 1", 1.25", 1.5", 1.75", 2"

a) Check wear: $w = \dfrac{KFNt}{3L}$ [Eq. 12-43]

$L > \dfrac{KFNt}{3w}$ for Oiles SP 500: $K = 0.6 \times 10^{-10}$ $\dfrac{in^3 \cdot min}{lb\ ft\ h}$

$$L > \frac{(0.6\times10^{-10}\ \frac{in^3\cdot min}{lb\cdot ft\cdot h})(100\ lb\cdot \underset{\downarrow}{n_d}\ 2)(200\ \frac{rev}{min})(1000\ h)}{3\,(0.002\ in)}$$

$L > 0.40$ in

Check Temperature:

$T_f \approx T_\infty + \dfrac{720\, f_s\, F\, n_d\, N}{J\, h_{CR}\, L}$ [Eq. 12-49]

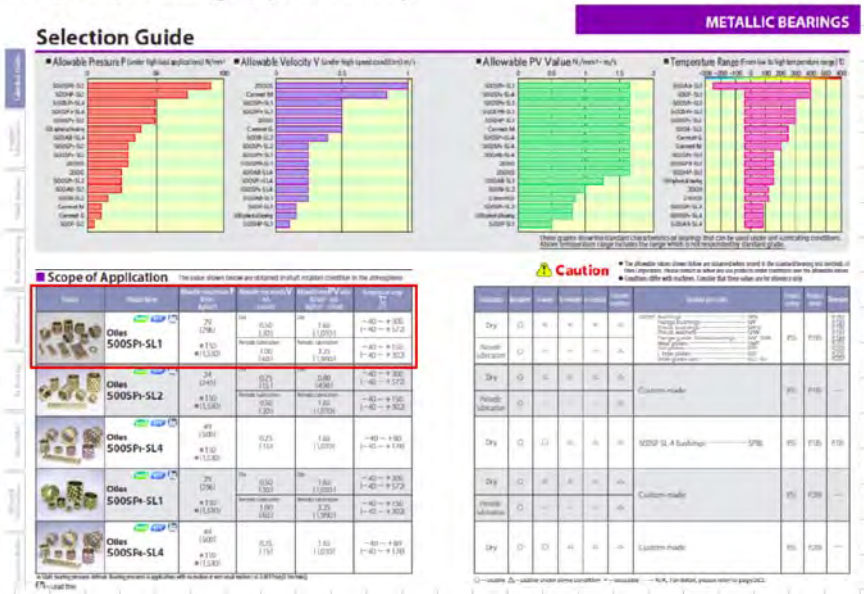
$L > \dfrac{720\, f_s\, F n_d N}{J\, h_{CR}\,(T_f - T_\infty)}$

$L > \dfrac{720\,(0.03)(100\cdot 2)(200)}{778\,(2.7)(300-70)}$ $L > 1.788$ in ← select greater length

ROUND UP TO AVAILABLE STANDARD SIZE: SELECT $L = 2.0$ in
ID = 1.25 in
OD = 1.625 in

$L/D = 1.6$ - OK

From Oiles Catalogue (See Canvas)



* $1 kgf/cm^2 = 14.2$ psi

$1 ft = 0.3048$ m

MAX PRESSURE : $296 kgf/cm^3 = 4203.2$ psi

$$\rightarrow P_{max} = \frac{4 F n_d}{\pi D L} = \frac{4(100 \cdot 2)}{\pi (1.25)(2.0)} = 101.9 \text{ psi} < 4203.2$$ - OK ✓

MAX VELOCITY : 30 m/min

$$\rightarrow V = \frac{\pi D N}{12} = \frac{\pi (1.25)(200)}{12} = 65.4 \frac{ft}{min} \times \frac{0.3048 m}{ft} = 19.95 \frac{m}{min}$$ - OK ✓

MAX PV : 1010 $kg/cm^3 \cdot m/min$

$$PV = \frac{\pi F n_d N}{12 L} = \frac{\pi (100 \cdot 2)(200)}{12 (2)} = 5236 \text{ psi} \cdot ft/min$$

$$5236 \frac{psi \cdot ft}{min} \times \frac{1 kgf/cm^2}{14.2 psi} \times \frac{0.3048 m}{ft} = 112.4 \frac{kgf \cdot m/min}{cm^2}$$ - OK ✓

TEMP RANGE : - 40 - 572 °F

→ 70 - 300 F within allowable range : OK ✓

CONFIRMED - MEETS ALL CATALOGUE LIMITS

Problem 3 (Shigley: Question 11-23)

An 02-series angular contact deep-groove ball bearing with a 30-mm bore (see Shigley Tables 11–1 and 11–2 for specifications) is loaded with a 2-kN axial load and a 5-kN radial load. The inner ring rotates at 400 rev/min.

(*a*) Determine the equivalent radial load that will be experienced by this particular bearing.

(*b*) Determine the predicted life (in revolutions) that this bearing could be expected to give in this application with a 99% reliability. (Use SKF Weibull parameters $x_o = 0.02$, $\theta = 4.459$, $b = 1.483$)

_11-23 (a) $F_a = 2$ kN, $F_r = 5$ kN, $n_D = 400$ rev/min, $V = 1$

From Table 11-2, 30 mm bore, $C_{10} = 19.5$ kN, $C_0 = 10.0$ kN

$$F_a / C_0 = 2 / 10 = 0.2$$

From Table 11-1, $0.34 \le e \le 0.38$.

$$\frac{F_a}{VF_r} = \frac{2}{(1)(5)} = 0.4$$

Since this is greater than e, interpolating Table 11-1, with $F_a / C_0 = 0.2$, we obtain $X_2 = 0.56$ and $Y_2 = 1.27$.

Eq. (11-12): $F_e = X_i V F_r + Y_i F_a = (0.56)(1)(5) + (1.27)(2) = 5.34$ kN *Ans.*

$F_e > F_r$ so use F_e.

(b) Solve Eq. (11-10) for x_D.

$$x_D = \left(\frac{C_{10}}{a_f F_D}\right)^a \left[x_0 + (\theta - x_0)(1 - R_D)^{1/b}\right]$$

$$x_D = \left(\frac{19.5}{(1)(5.34)}\right)^3 \left[0.02 + (4.459 - 0.02)(1 - 0.99)^{1/1.483}\right]$$

$$x_D = 10.66$$

$$x_D = \frac{L_D}{L_R} = \frac{L_D n_D (60)}{10^6}$$

$$L_D = \frac{x_D (10^6)}{n_D (60)} = \frac{10.66(10^6)}{(400)(60)} = 444 \text{ h} \quad \textit{Ans.}$$

a) Equivalent Load:

$F_a = 2\ kN$
$F_r = 5\ kN$

BORE = 30 mm

- use ANGULAR CONTACT RATINGS:

$C_o = 11.0\ kN$
$C_{10} = 20.3\ kN$

Table 11-2

Dimensions and Load Ratings for Single-Row 02-Series Deep-Groove and Angular-Contact Ball Bearings

Bore, mm	OD, mm	Width, mm	Fillet Radius, mm	Shoulder Diameter, mm d_s	Shoulder Diameter, mm d_H	Load Ratings, kN Deep Groove C_{10}	Deep Groove C_o	Angular Contact C_{10}	Angular Contact C_o
10	30	9	0.6	12.5	27	5.07	2.24	4.94	2.12
12	32	10	0.6	14.5	28	6.89	3.10	7.02	3.05
15	35	11	0.6	17.5	31	7.80	3.55	8.06	3.65
17	40	12	0.6	19.5	34	9.56	4.50	9.95	4.75
20	47	14	1.0	25	41	12.7	6.20	13.3	6.55
25	52	15	1.0	30	47	14.0	6.95	14.8	7.65
30	62	16	1.0	35	55	19.5	10.0	20.3	11.0
35	72	17	1.0	41	65	25.5	13.7	27.0	15.0
40	80	18	1.0	46	72	30.7	16.6	31.9	18.6

$$F_e = XVF_r + YF_a \quad [Eq.\ 11\text{-}12]$$

$$F_a/C_o = \frac{2\ kN}{11\ kN} = 0.18182$$

[Table 11-1]

F_a/C_o	e	$F_a/(VF_r) \le e$ X_1	$F_a/(VF_r) \le e$ Y_1	$F_a/(VF_r) > e$ X_2	$F_a/(VF_r) > e$ Y_2
0.014*	0.19	1.00	0	0.56	2.30
0.021	0.21	1.00	0	0.56	2.15
0.028	0.22	1.00	0	0.56	1.99
0.042	0.24	1.00	0	0.56	1.85
0.056	0.26	1.00	0	0.56	1.71
0.070	0.27	1.00	0	0.56	1.63
0.084	0.28	1.00	0	0.56	1.55
0.110	0.30	1.00	0	0.56	1.45
0.17	0.34	1.00	0	0.56	1.31
0.28	0.38	1.00	0	0.56	1.15
0.42	0.42	1.00	0	0.56	1.04
0.56	0.44	1.00	0	0.56	1.00

*Use 0.014 if $F_a/C_o < 0.014$.

Interpolate between 0.17 - 0.28

$$e = \frac{(0.18182 - 0.17)}{(0.28 - 0.17)} \times (0.38 - 0.34) + 0.34$$

$$e = 0.3443$$

interpolate

$$\frac{F_a}{VF_r} = \frac{2\ kN}{(1.0)(5\ kN)} = 0.4 > e$$

(V = 1 : inner ring rotates)

use $X_2 = 0.56$

$$Y_2 = 1.31 - \left(\frac{0.182 - 0.17}{0.28 - 0.17}\right) \times (1.31 - 1.15)$$

interpolate to get $Y_2 = 1.293$

$$F_e = 0.56\,(1.0)(5\ kN) + 1.293\,(2\ kN)$$

$$F_e = 5.386\ kN$$

b) Predicted Life :

$N = 400$ rpm
$R = 0.99$

$x_o = 0.02$
$\theta = 4.459$
$b = 1.483$

$L_{10} = 10^6$ revolutions

$F_e = 5.386$ kN (- assume $a_f = 1.0$)
$C_{10} = 20.3$ kN
$a = 3.0$ for ball bearings

[Eq. 11-9]

$$C_{10} = a_f F_e \left[\frac{x_D}{x_o + (\theta - x_o)\left[\ln\left(\frac{1}{R}\right)\right]^{1/b}} \right]^{1/a} \rightarrow \text{Solve for } x_D$$

$$\left(\frac{C_{10}}{a_f F_e}\right)^3 \left(x_o + (\theta - x_o)\left[\ln\left(\frac{1}{R}\right)\right]^{1/b} \right) = x_D$$

$$x_D = \left(\frac{20.3}{1 \cdot 5.386}\right)^3 \left(0.02 + (4.459 - 0.02)\left[\ln\left(\frac{1}{0.99}\right)\right]^{1/1.483}\right)$$

$$x_D = 11.76 = \frac{L_D}{L_{10}}$$

$$L_D = 11.76 \times 10^6 \text{ cycles} \approx 11\,759\,600 \text{ revolutions}$$

PREDICTED LIFE : $L_D \approx 11.76 \times 10^6$ revolutions

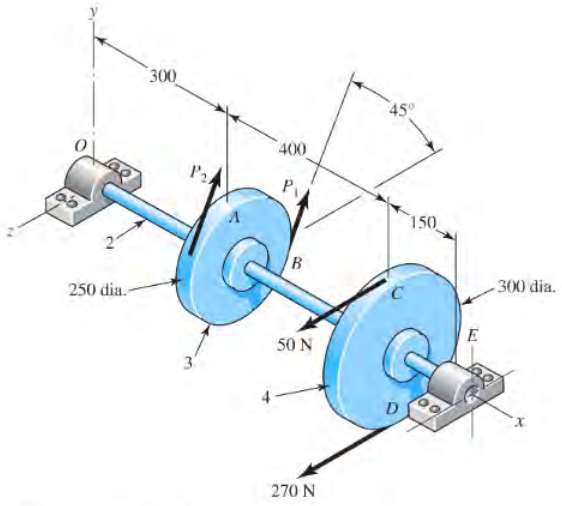
(@ 400 rpm

11.76×10^6 rev $\times \frac{1 \text{ min}}{400 \text{ rev}} \times \frac{1 \text{ h}}{60 \text{ min}} \approx 490$ hours)

Problem 4 (Shigley: Problem 11-35 Modified)

A countershaft supports two V-belt pulleys. The shaft is operating at 1500 rpm. The belt tension on the loose (slack) side of pulley A is 15% of the tight (taut) side. The bearings are expected to operate for 60000 hours with a **combined** reliability of 98%. (Use SKF Weibull parameters $x_o = 0.02$, $\theta = 4.459$, $b = 1.483$)

a) Select deep groove ball bearings from Table 11-2 for use at O and E. Use an application factor of 1.0.
b) Select appropriate bearings for O and E using the SKF Bearing Catalogue.
c) Determine the tolerance bands for the SKF bearings selected in part b.

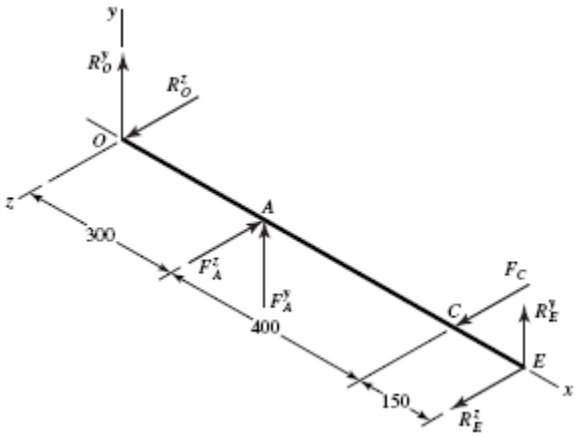


Problem 11–35

Dimensions in millimeters.

11-35 The reliability of the individual bearings is $R = \sqrt{0.98} = 0.9899$
From statics,

$T = (270 - 50) = (P_1 - P_2)125$
$= (P_1 - 0.15\,P_1)125$
$P_1 = 310.6$ N,
$P_2 = 0.15\,(310.6) = 46.6$ N
$P_1 + P_2 = 357.2$ N
$F_A^y = 357.2 \sin 45^\circ = 252.6 \text{ N} = F_A^z$



$$\sum M_O^z = 850R_E^y + 300(252.6) = 0 \Rightarrow R_E^y = -89.2 \text{ N}$$
$$\sum F^y = 252.6 - 89.2 + R_O^y = 0 \Rightarrow R_O^y = -163.4 \text{ N}$$

$$\sum M_O^y = -850R_E^z - 700(320) + 300(252.6) = 0 \Rightarrow R_E^z = -174.4 \text{ N}$$
$$\sum F^z = -174.4 + 320 - 252.6 + R_O^z = 0 \Rightarrow R_O^z = 107 \text{ N}$$

$$R_O = \sqrt{(-163.4)^2 + 107^2} = 195 \text{ N}$$
$$R_E = \sqrt{(-89.2)^2 + (-174.4)^2} = 196 \text{ N}$$

The radial loads are nearly the same at *O* and *E*. We can use the same bearing at both locations.

$$x_D = \frac{60\,000(1500)(60)}{10^6} = 5400$$

Eq. (11-9): $$C_{10} = 1(0.196)\left\{\frac{5400}{0.02 + 4.439\left[\ln(1/0.9899)\right]^{1/1.483}}\right\}^{1/3} = 5.7 \text{ kN}$$

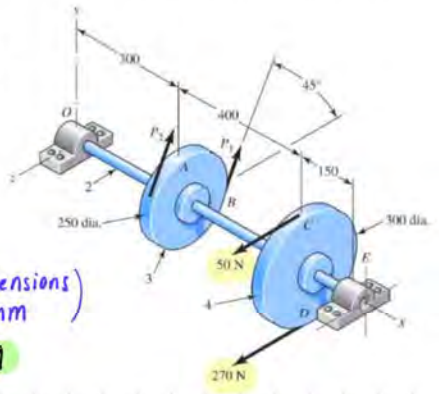
From Table 11-2, select an 02-12 mm deep-groove ball bearing with a basic load rating of 6.89 kN. *Ans.*

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4) **Problem 4 (Shigley: Problem 11-35 Modified)**

A countershaft supports two V-belt pulleys. The shaft is operating at 1500 rpm. The belt tension on the loose (slack) side of pulley A is 15% of the tight (taut) side. The bearings are expected to operate for 60000 hours with a **combined** reliability of 98%. (Use SKF Weibull parameters $x_o = 0.02$, $\theta = 4.459$, $b = 1.483$)

a) Select deep groove ball bearings from Table 11-2 for use at O and E. Use an application factor of 1.0.
b) Select appropriate bearings for O and E using the SKF Bearing Catalogue.
c) Determine the tolerance bands for the SKF bearings selected in part b.



a)

$N = 1500$ rpm $a_f = 1.0$ (* dimensions in mm)

$R_{comb} = 0.98$ * COMBINED RELIABILITY $R = \sqrt{0.98} = 0.99$

$L_D = 60\,000\ h \times \frac{60\ min}{h} \times \frac{1500\ rev}{min} = 5400 \times 10^6$ revolutions

$x_D = \frac{L_D}{L_{10}} = 5400$

@ PULLEY A, $P_2 = 0.15\ P_1$

· equal torque along shaft: $+\circlearrowleft \Sigma M_x = 0$ at steady state

$$+270\,N(0.15\,m) - 50(0.15) - P_1(0.125) + P_2(0.125) = 0$$

$P_2 = 0.15\ P_1$

$$33\ N\text{-}m = 0.85\ P_1\ (0.125)$$

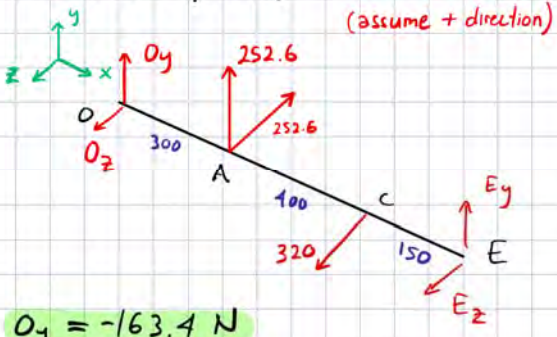
$P_1 = 310.6$ N
$P_2 = 46.6$ N

$P_T = P_1 + P_2 = 357.2\ N\ @\ 45°$

$A_y = 357.2 \sin 45° = 252.6\ N\ \uparrow$
$A_z = 357.2 \cos 45° = 252.6\ N\ \rightarrow$

$C_z = -320\ N\ (\swarrow)$

Find Radial Loads at Bearings O and E (no thrust loads present)



(assume + direction)

$\overset{+}{\uparrow}\overset{y}{\rightarrow} \Sigma M_O = 252.6(300) - 320(700) - E_z(850) = 0$

$E_z = -174.4\ N$

$+\circlearrowleft \Sigma M_O = 252.6(300) + E_y(850) = 0$

$E_y = -89.15\ N$

$+\uparrow \Sigma F_y = O_y + 252.6 - 89.15 = 0 \rightarrow O_y = -163.4\ N$

$\overset{+}{\swarrow} \Sigma F_z = O_z - 252.6 + 320 - 174.4 \rightarrow O_z = +107\ N$

$|E_r| = \sqrt{(174.4)^2 + (89.15)^2} = 195.86\ N$

$|O_r| = \sqrt{163.4^2 + 107^2} = 195.3\ N$

Size both bearings for 196 N RADIAL LOAD

$$C_{10} = a_f F_D \left[\frac{X_D}{0.02 + 4.439 \ln\left(\frac{1}{0.99}\right)^{1/1.483}} \right]^{1/3}$$

$$= 1\,(196\ N) \left[\frac{5400}{0.2196} \right]^{1/3} = 5699.7\ N$$

$$= 5.7\ kN$$

[Table 11-2] : Select Deep Groove :

02 series, 12 mm bore, 6.89 kN rating for both bearings

b) SKF [catalog. pg. 268]

Select 15 mm bore, 35 mm OD, 8.06 kN rating ‖ 6202

c) TOLERANCE BANDS

Rotating inner / Stationary outer ring
Constant Load direction
} INTERFERENCE FIT INNER / LOOSE FIT OUTER

[SKF Inst / Maintenance Guide] - Table 2 (pg. 54)

$$P/C = \frac{196}{8060} = 0.024 \rightarrow P < 0.05C$$

Shaft < 17 mm : js5 tolerance

Table 5 : Housing H8

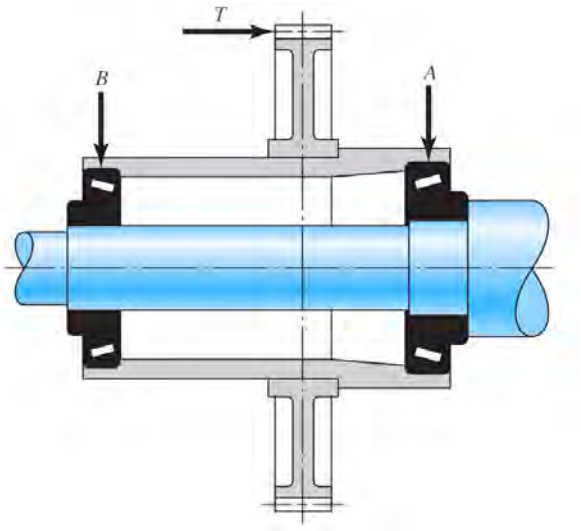
Table 8 : (pg. 62)
Shaft max : 0.5908 in
min : 0.5904 in
for 15 mm nominal

Table 9 : (pg. 72)
Housing min : 1.3780 in
max : 1.3795 in
for 35 mm nominal OD

Problem 5 (Shigley 11-50)

The gear reduction unit shown has a gear that is press fit onto a cylindrical sleeve that rotates around a stationary shaft. The helical gear transmits an axial thrust load, T, of 250lbf. Tangential and radial loads (not shown) are also transmitted through the gear, producing radial forces of 875 lbf at bearing A and 625 lbf. at bearing B. The desired life for each bearing is 90000 hours at a speed of 150 rpm with 90% reliability. The first iteration of the shaft design indicates approximate diameters of 1.125 inch at A and 1.000 inch at B.

Select appropriate Timken roller bearings for the shaft. (Use Timken Weibull parameters x_o = 0.0, θ = 4.48, b = 1.50)



11-50 The left bearing is compressed by the axial load, so it is properly designated as bearing *A*.

$$F_{rA} = 875 \text{ lbf}$$
$$F_{rB} = 625 \text{ lbf}$$
$$F_{ae} = 250 \text{ lbf}$$

Assume K = 1.5 for each bearing for the first iteration. Obtain the induced loads.

Eq. (11-18): $$F_{iA} = \frac{0.47 F_{rA}}{K_A} = \frac{0.47(875)}{1.5} = 274 \text{ lbf}$$

Eq. (11-18): $$F_{iB} = \frac{0.47 F_{rB}}{K_B} = \frac{0.47(625)}{1.5} = 196 \text{ lbf}$$

Check the condition on whether to apply Eq. (11-19) or Eq. (11-20).

$$F_{iA} \le ? \ge F_{iB} + F_{ae}$$
$$274 \text{ lbf} < 196 + 250 \text{ lbf, so Eq.(11-19) applies}$$

We will size bearing *B* first since its induced load will affect bearing *A*, but it is not affected by the induced load from bearing *A* [see Eq. (11-19)].

From Eq. (11-19*b*), $F_{eB} = F_{rB} = 625$ lbf.

Eq. (11-3): $$F_{RB} = a_f F_D \left[\frac{L_D}{L_R}\right]^{3/10} = 1(625)\left[\frac{90\,000(150)(60)}{90(10^6)}\right]^{3/10}$$

$$F_{RB} = 1208 \text{ lbf}$$

Select cone 07100, cup 07196, with 1 in bore, and rated at 1570 lbf with $K = 1.45$. *Ans.*

With bearing *B* selected, we re-evaluate the induced load from bearing *B* using the actual value for *K*.

Eq. (11-18): $$F_{iB} = \frac{0.47F_{rB}}{K_B} = \frac{0.47(625)}{1.45} = 203 \text{ lbf}$$

Find the equivalent radial load for bearing *A* from Eq. (11-19), which still applies.

Eq. (11-19*a*): $$F_{eA} = 0.4F_{rA} + K_A(F_{iB} + F_{ae})$$

$$= 0.4(875) + 1.5(203 + 250) = 1030 \text{ lbf}$$

$$F_{eA} > F_{rA}$$

Eq. (11-3): $$F_{RA} = a_f F_D \left[\frac{L_D}{L_R}\right]^{3/10} = 1(1030)\left[\frac{90\,000(150)(60)}{90(10^6)}\right]^{3/10}$$

$$F_{RA} = 1990 \text{ lbf}$$

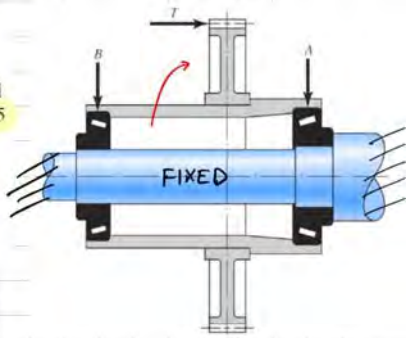
Any of the bearings with 1-1/8 in bore are more than adequate. Select cone 15590, cup 15520, rated at 2480 lbf with $K = 1.69$. Iterating with the new value for *K*, we get $F_{eA} = 1120$ lbf and $F_{rA} = 2160$ lbf. The selected bearing is still adequate. *Ans.*

5)

Problem 5 (Shigley 11-50)

The gear reduction unit shown has a gear that is press fit onto a cylindrical sleeve that rotates around a stationary shaft. The helical gear transmits an axial thrust load, T, of 250lbf. Tangential and radial loads (not shown) are also transmitted through the gear, producing radial forces of 875 lbf at bearing A and 625 lbf. at bearing B. The desired life for each bearing is 90000 hours at a speed of 150 rpm with 90% reliability. The first iteration of the shaft design indicates approximate diameters of 1.125 inch at A and 1.000 inch at B.

Select appropriate Timken roller bearings for the shaft. (Use Timken Weibull parameters x_o = 0.0, θ = 4.48, b = 1.50)



$F_a = 250$ lb (carried by bearing A)

$F_r = 875$ lb (A) $\quad F_r = 625$ lb (B) - assume $a_f = 1.0$

$L_D = 90\,000$ hr @ $N = 150$ rpm

$L_D = 90\,000\ hr \times \frac{60\ min}{hr} \times \frac{150\ rev}{min} = 810 \times 10^6$ cycles $\quad R = 0.90$

TIMKEN:

$x_o = 0$
$\theta = 4.48$
$b = 1.5$

$L_{10} = 90 \times 10^6 \rightarrow \boxed{x_D = \frac{810}{90} = 9.0}$

BEARING A:

[Eq. 11-19] $F_i = \frac{0.47\ F_r}{K}$ - assume $K = 1.5$ for first iteration

$F_{iA} = \frac{0.47(875)}{1.5} = 274.2$ lb

$F_e = 0.4F_r + KF_a$

$F_{eA} = 0.4(875) + 1.5(250)$

$F_{eA} = 725$ lbf

BEARING B:

$F_{iB} = \frac{0.47(625)}{1.5}$

$F_{iB} = 195.83$ lb

$F_{eB} = F_r = 625$ lb

Check F_{iA} vs. $(F_{iB} + F_{ae})$

$274.2 \leq (195.83 + 250)$ → use [Eq. 11-19 a/b]

$F_{eA} = 0.4\,F_{rA} + K_A\,(F_{iB} + F_{ae})$ // $F_{eB} = F_{rB} = 625$ lb

$= 0.4(875) + 1.5\,(195.8 + 250)$

$F_{eA} = 1018.75$ lb

Eq. 11-3 Select Bearing B:

$C_{10} = a_f\,F_D\,(x_D)^{1/a} = 1.0\,(625)(9)^{3/10}$

$= 1208.2$ lb

Select 1" BORE, 1570 lb [07100 / 07196] @ B
CONE CUP

$K = 1.45$

↳ update for bearing A selection

$$F_{iB} = \frac{0.47(625)}{1.45} = 202.59 \text{ lb}$$

$$F_{eA} = 0.4(875) + 1.5(202.6 + 250) = 1028.9 \text{ lb}$$

$$C_{10} = 1.0(1028.9)(9)^{3/10} = 1989.0 \text{ lb}$$

Select 1.125" BORE, 2480 lb [15590 / 15520] @ A
CONE CUP

$K = 1.69$

↳ Check with updated K

$$F_{eA} = 0.4(875) + 1.69(202.6 + 250) = 1115 \text{ lb}$$

$$C_{10} = 1.0(1115)\,9^{3/10} = 2155 \text{ lb}$$ ✓ OK

Selected bearing is sufficient.

	CONE	CUP
A:	15590	/ 15520
B:	07100	/ 07196

Problem 6 (Mott: Problem 11-4 Modified)

A chain sprocket delivers 110 HP to a shaft at a rotational speed of 1700 rpm. The sprocket has a bore diameter of 2.5 inches. The hub length is 3.25 inches.

a) Specify a suitable key for this application. You can assume the sprocket is made of SAE 4140 OQT 1300 and the key is SAE 1018 (Table 11-4).
b) Specify a suitable spline for the application with a B fit.
c) Specify a suitable pin for the application.
d) Which one do you recommend? Why?

Nominal Pin Diameter,[2] in	Nominal Failure Load in Double Shear,[1] lb					
	Clevis[3]	Dowel[4]	Taper[3]	Grooved[4]	Spring[4]	Spiral, Heavy Duty[4]
0.031	—	—	—	200	—	—
0.062	—	800	700	780	425	450
0.125	—	3200	2900	3120	2100	2100
0.188	6400	7200	6600	6990	4400	4400
0.250	11,330	12,800	11,800	12,430	7200	7700
0.375	25,490	28,700	26,400	27,950	17,600	17,600
0.500	45,320	51,000	46,900	49,700	25,800	30,000
0.625	70,810	79,800	73,400	—	46,000	46,000
0.750	101,970	114,000	104,900	—	66,000	66,000
0.875	138,780	156,000	143,500	—	—	—
1.000	181,270	204,000	187,700	—	—	—

[1]Tangential force at shaft–hub interface that will shear off the pin assuming the load to be equally distributed between the two shear areas.
[2]Other standard sizes available.
[3]Based on 1095 steel quenched and drawn to Rockwell C 42 (see Table 3.3). Other materials are available.
[4]Selected data excerpted from ref. 12.

Part a)

From Table 11-4, for SAE 1018 steel, we have $s_y =$ 54000 psi and $s_u =$ 64000 psi. From Appendix for SAE 4140 OQT 1300 we have and $s_y =$ 100000 psi and $s_u =$ 117000 psi.

Typically a design factor of N=3 is used and we apply the Trseca criteria for shear ($\tau_y = 0.5 s_y$) is so that $\tau_d = 0.5(54000)/N=3 = 9000$psi

$T = (63\,000)(110)/1700 = 4076\ \text{lb} \cdot \text{in};\ D_s = 2.50\ \text{in};\ \text{Use}\ \frac{5}{8}\ \text{square key.}$

$$L = \frac{2T}{\tau_d DW} = \frac{2(4076)}{(9000)(2.50)(0.625)} = 0.580\ \text{in}$$

Use $L = 2.50$ in to more nearly match hub length.

Hub length is 3.25" so L = 3.25"

Part b)

Express data from Table 11-6 as $T = KD^2L$

Required $K = T/D^2L$ (L = Hub length) Use B – Fit

Problem 4 data: $T = 4076\ \text{lb} \cdot \text{in};\ D = 2.50\ \text{in};\ L = 3.25\ \text{in}$

$$K = \frac{4076}{(2.50)^2(3.25)} = \mathbf{201};\ \textbf{Use either 4 } (\boldsymbol{K} \textbf{ = 219}) \textbf{ or 6 splines } (\boldsymbol{K} \textbf{ = 208})$$

If Safety factor is used, we will increase the number of splines to 16 (see below)

6)

Problem 6 (Mott: Problem 11-4 Modified)

A chain sprocket delivers 110 HP to a shaft at a rotational speed of 1700 rpm. The sprocket has a bore diameter of 2.5 inches. The hub length is 3.25 inches.

a) Specify a suitable key for this application. You can assume the sprocket is made of SAE 4140 OQT 1300 and the key is SAE 1018 (Table 11-4).
b) Specify a suitable spline for the application with a B fit.
c) Specify a suitable pin for the application.
d) Which one do you recommend? Why?

a) SPROCKET: SAE 4140 QQT 1300 $\begin{cases} S_u = 117 \text{ ksi} \\ S_y = 100 \text{ ksi} \end{cases}$ [Table 11-4, pg. 476]

KEY: SAE 1018 $\begin{cases} S_u = 64 \text{ ksi} \\ S_y = 54 \text{ ksi} \end{cases}$ [Appendix 3 - pg. 787] - Note KEY has lower yield strength

$$T = \frac{63025\,H}{n} = \frac{63025\,(110\text{ hp})}{1700\text{ rpm}} = 4078.1 \text{ lb·in} \rightarrow F = 3262.5 \text{ lb}$$

$d = 2.5''$ (1.25 in)

Select $W = 5/8''$ / $H = 5/8''$ [See Shigley, Table 7-6, pg. 403]

Check Shear: Use $N = 3.0$ [See pg. 477]

$$\tau_d = \frac{0.5\,S_y}{N} = \frac{54 \text{ ksi}/2}{3.0} = 9.0 \text{ ksi}$$

$$L_{min} = \frac{2T}{\tau_d D W} = \frac{2\,(4078.1 \text{ lb-in})}{9000\text{psi}\,(2.5\text{in})(5/8\text{ in})} \quad \text{[Eq. 11-2]} = 0.58 \text{ in}$$

$\frac{F}{WL} < S_y/3$

$\frac{3F}{WL} < S_y$

Check Compression:

$$\sigma_d = \frac{S_y}{N} = \frac{54 \text{ ksi}}{3} = 18.0 \text{ ksi}$$

$$L_{min} = \frac{4T}{\sigma_d D H} = \frac{4\,(4078.1 \text{ lb·in})}{18000\text{ psi}\,(2.5\text{in})(5/8\text{ in})} = 0.58 \text{ in}$$

$T = \frac{Fd}{2}$ $\frac{3T \cdot 2}{dWL} < S_y$

Mott suggests key should extend over all of hub length [See pg. 477]

→ Select 3.25" Length × 5/8" H × 5/8" W

b) Spline, B fit

[Table 11-5]

TABLE 11–5 Formulas for SAE Straight Splines

Number of splines	W, for all fits	A: Permanent fit h	A: Permanent fit d	B: To slide without load h	B: To slide without load d	C: To slide under load h	C: To slide under load d
Four	0.241D	0.075D	0.850D	0.125D	0.750D		
Six	0.250D	0.050D	0.900D	0.075D	0.850D	0.100D	0.800D
Ten	0.156D	0.045D	0.910D	0.070D	0.860D	0.095D	0.810D

- assume spline length = hub length = 3.25 in
- Select 6 splines

[Table 11-6]

TABLE 11-6 Torque Capacity for Straight Splines per Inch of Spline Length

Number of splines	Fit	Torque capacity	Required diameter
4	A	$139D^2$	$\sqrt{T/139}$
4	B	$219D^2$	$\sqrt{T/219}$
6	A	$143D^2$	$\sqrt{T/143}$
6	B	$208D^2$	$\sqrt{T/208}$
6	C	$270D^2$	$\sqrt{T/270}$

① using $d = 2.5$" (hub size of sprocket)

→ $d = 0.85D$ → $D = 2.95$ in [6B]

$h = 0.075D$ → $h = 0.221$ in

Torque capacity/in $= 200D^2$

$= 1799.3$ lb·in/in × 3.25 in length

$T = 5847.8$ in-lb $\quad SF = \frac{5847}{4078} = 1.4$

× not sufficient

- SF 3.0 recommended

② Increase # of splines to 16 B

$d = 0.86D$ → $D = 2.907$ in

$h = 0.07D$ → $h = 0.203$ in

$W = 0.098D$ → $W = 0.285$ in

16 splines

$T = 521D^2$

$= 521(2.907\text{ in})^2 = 4402.7$ lb in/in capacity

@ 3.25 in length : $T = 14\,308.8$ lb-in

$SF = 3.51$ ✓ OK

c) Pin $F = 3262.5$ lb @ $r = 1.25$ in

(Double Shear : $F = T/D = \frac{4078\text{ lb-in}}{2.5\text{ in}} = 1631.2$ lb @ each end of pin)

using $N = 3$, $F_{req} = 3(3262.5\text{ lb}) = 9787.4$ lb req

Select Dowel pin :

0.250" ø

Nominal Failure Load in Double Shear,[1] lb

Nominal Pin Diameter,[2] in	Clevis[3]	Dowel[4]	Taper[3]	Grooved[4]	Spring[4]	Spiral, Heavy Duty[4]
0.031	—	—	—	200	—	—
0.062	—	800	700	780	425	450
0.125	—	3200	2900	3120	2100	2100
0.188	6400	7200	6600	6990	4400	4400
0.250	11,330	12,800	11,800	12,430	7200	7700
0.375	25,490	28,700	26,400	27,950	17,600	17,600
0.500	45,320	51,000	46,900	49,700	25,800	30,000
0.625	70,810	79,800	73,400	—	46,000	46,000
0.750	101,970	114,000	104,900	—	66,000	66,000
0.875	138,780	156,000	143,500	—	—	—
1.000	181,270	204,000	187,700	—	—	—

[1]Tangential force at shaft–hub interface that will shear off the pin assuming the load to be equally distributed between the two shear areas.
[2]Other standard sizes available.
[3]Based on 1095 steel quenched and drawn to Rockwell C 42 (see Table 3.3). Other materials are available.
[4]Selected data excerpted from ref. 12.

d) Recommendation : PIN, due to simplicity and lowest cost (- splines and keys require more complex machining)

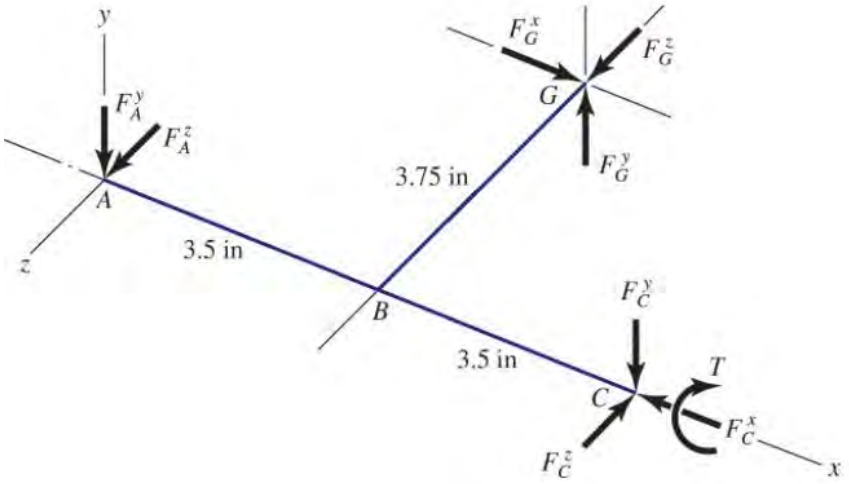
- can be downsized to provide "mechanical fuse" to shear if chain becomes jammed to protect other components of the system

Part d) Of the three options, the spline will be most expensive to manufacture so it is not recommended. The key is the most common method for delivering torque and with a length equal to the hub length, it is a good choice. The pin is also a suitable choice and will be a suitable mechanical fuse. A taper pin has little chance to fall out of the hole in this application. Hence, a key or pin are both suitable.

Problem 7 (Shigley: Problem 7-6 Modified)

A set of bevel gears transmit torsional, bending, and axial loads to rotating shafts. For this problem, the free body diagram for a shaft *ABC* is shown below, where points *A* and *C* are at the bearings. Line *BG* is on the bevel gear extended to the shaft center from point *G*, the contact point with the mating gear. Transmitted from the mating gear are tangential, radial, and axial forces FGy=500 lbf (tangential), FGz=58 lbf (radial) and FGx=173 lbf (axial), respectively. The bearing at *C* supports the axial thrust load. For the shaft material, Su=90 kpsi, Sy=65 kpsi, and has a fully corrected endurance limit of 40 kpsi.

Let K_{fa} =2.3, K_{fb} =2.0, and K_{fst} =1.5, where the subscripts a, b, and t represent axial, bending, and torsion, respectively. The diameter of shaft *ABC* is 1.5 in.



Using the Von-misses equations for a shaft (Eq's 6-66 and 6-67) and the alternating and mean stress equations (Eq'ns 7-2, 7-3) one can show that the design equation for the safety factor, n_f, using the Distortion Energy (DE) Goodman failure criteria is:

$$n_f = \frac{\pi d^3}{4}\left\{\frac{1}{S_e}\left[\left(8K_{fb}M_a + K_{fa}F_a d\right)^2 + 48\left(K_{fs}T_a\right)^2\right]^{1/2} + \frac{1}{S_{ut}}\left[\left(8K_{fb}M_m + K_{fa}F_m d\right)^2 + 48\left(K_{fs}T_m\right)^2\right]^{1/2}\right\}^{-1}$$

a) Determine the value of the design factor for the bevel gears.
b) Determine the value of the design factor ignoring the axial force. What is the contribution from the axial force? Is it important?

7-6 **(a)** Let $K_{fb} = (K_f)_{\text{bending}}$, $K_{fa} = (K_f)_{\text{axial}}$, $K_{fs} = (K_{fs})_{\text{torsion}}$.
Eqs. (6-66) and (6-67):

$$\sigma_a' = \left[\left(K_{fb} \frac{32M_a}{\pi d^3} + K_{fa} \frac{4F_a}{\pi d^2} \right)^2 + 3\left(K_{fs} \frac{16T_a}{\pi d^3} \right) \right]^{1/2}$$

$$\sigma_m' = \left[\left(K_{fb} \frac{32M_m}{\pi d^3} + K_{fa} \frac{4F_m}{\pi d^2} \right)^2 + 3\left(K_{fs} \frac{16T_m}{\pi d^3} \right) \right]^{1/2}$$

For the DE-Goodman criterion, substitute stresses into Eq. (6-41),

$$n_f = \frac{\pi d^3}{4} \left\{ \begin{array}{l} \dfrac{1}{S_e} \left[\left(8K_{fb}M_a + K_{fa}F_a d \right)^2 + 48\left(K_{fs}T_a \right)^2 \right]^{1/2} \\ + \dfrac{1}{S_{ut}} \left[\left(8K_{fb}M_m + K_{fa}F_m d \right)^2 + 48\left(K_{fs}T_m \right)^2 \right]^{1/2} \end{array} \right\}^{-1} \quad \textit{Ans.}$$

(b) For the free-body diagram in the problem statement,

$$\sum F_x = 0 \quad \Rightarrow \quad F_C^x = 173 \text{ lbf}$$

$$\left(\sum M_A \right)_y = 0 = 7F_C^z - 3.75(173) - 3.5(58) \quad \Rightarrow \quad F_C^z = 121.68 \text{ lbf}$$

$$\left(\sum M_A \right)_z = 0 = -7F_C^y + 3.5(500) \quad \Rightarrow \quad F_C^y = 250 \text{ lbf}$$

$$\left(\sum T_A \right)_x = 0 = 3.75(500) - T \quad \Rightarrow \quad T = 1875 \text{ lbf} \cdot \text{in}$$

$$M_m = 0, \qquad M_a = M_B = 3.5\sqrt{121.68^2 + 250^2} = 973.1 \text{ lbf} \cdot \text{in}$$

$T_m = 1875$ lbf · in, $T_a = 0$, $F_m = 173$ lbf, $F_a = 0$
From part (a), with $d = 1.5$ in,

$$n_f = \frac{\pi(1.5^3)}{4} \left\{ \begin{array}{l} \dfrac{1}{40(10^3)} \left\{ \left[8(2)(973.1) + 0 \right]^2 \right\}^{1/2} \\ + \dfrac{1}{90(10^3)} \left\{ \left[2.3(173)1.5 \right]^2 + 48\left[1.5(1875) \right]^2 \right\}^{1/2} \end{array} \right\}^{-1} = 4.38 \quad \textit{Ans.}$$

(c) Without axial force, use Eq. (7-7):

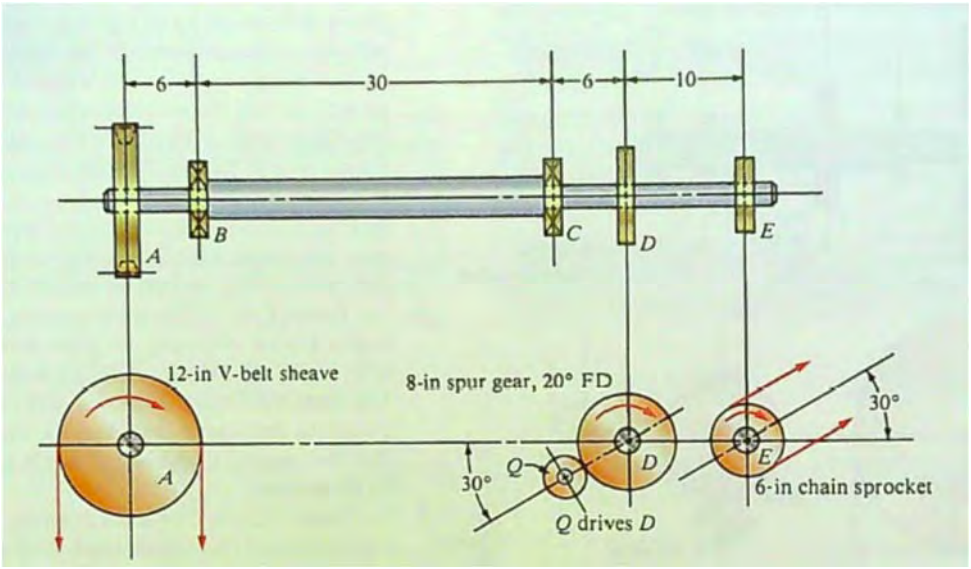
$$\frac{1}{n} = \frac{16}{\pi(1.5^3)} \left\{ \frac{1}{40(10^3)} \left\{ 4\left[2(973.1) \right]^2 \right\}^{1/2} + \frac{1}{90(10^3)} \left\{ 3\left[1.5(1875) \right]^2 \right\}^{1/2} \right\} = 4.38 \quad \textit{Ans.}$$

It is important to recognize from this example that the axial force is too small to add any appreciable difference. Hence it can ignored in most shaft analysis design problems.

Problem 8 (Mott: Problem 12-26 Modified)

A shaft is part of a conveyor system designed to deliver crushed rock to railway cars.

- The shaft rotates at 240 rpm
- A pinion gear at Q delivers 15HP to gear D at an angle of -30°
- The spur gear at D has 48 teeth with diametrical pitch of 6 teeth/inch. The teeth are 20° full-depth teeth.
- The 12 inch diameter V-belt sheave at A delivers 10HP to a mating sheave directly below
- The 6 inch chain drive sprocket at E delivers 5HP at an angle of 30°
- The torque at A, D, and E are delivered through keys (Tables 11-1 &11-4).
- The sprocket, gear and sheaves (A, D, and E) are held in place with retaining rings on both sides (no shoulders to position these components). Assume that the gear, sprockets, and sheaves have hubs that can be bored to appropriate diameters.
- Bearings at B and C are pressed against shoulders with sharp fillets
- The shaft is made from SAE 1137 Cold Drawn (CD) steel.
- The shaft is expected to operate 8 hours/day, 20 days/month for 5 years.



a) Determine the reactions at the Roller Bearing locations B and C. Ignore the weight of the shaft.
b) Complete the force, shear, bending moment and torque diagrams for the shaft.
c) Determine the minimum shaft diameter for i) chain sprocket, ii) V-belt sheave, and iii) the spur gear. Use safety factor of 3 and 99% reliability
d) Size appropriate keys for the components.
e) Size appropriate deep groove ball bearings for locations B and C (assume 90% reliability for the bearings). Be sure to specify the appropriate shoulder diameters.

26. Figure P12-5

Torque on Sheave A: $T_A = 63\,000(10)/240 = 2625$ lb · in $= T_{AD}$ In shaft

Torque on Gear D: $T_D = 63\,000(15)/240 = 3938$ lb · in

Torque on Sprocket E: $T_E = 63\,000(5)/240 = 1313$ lb · in $= T_{DE}$ In shaft

Forces on Sheave A:

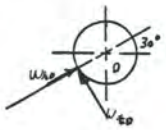
$$F_1 - F_2 = \frac{T_A}{R_A} = \frac{2625}{6} = 438 \text{ lb}; \quad F_{A_x} = 0$$

$$F_A = F_{A_X} = F_1 + F_2 = 1.5(F_1 - F_2) = 1.5(438) = 657 \text{ lb} \downarrow$$

Forces on Gear D:

$$W_{t_D} = \frac{T_D}{R_D} = \frac{3938}{4} = 985 \text{ lb}$$

$$W_{r_D} = W_{t_D} \tan 20° = 358 \text{ lb}$$



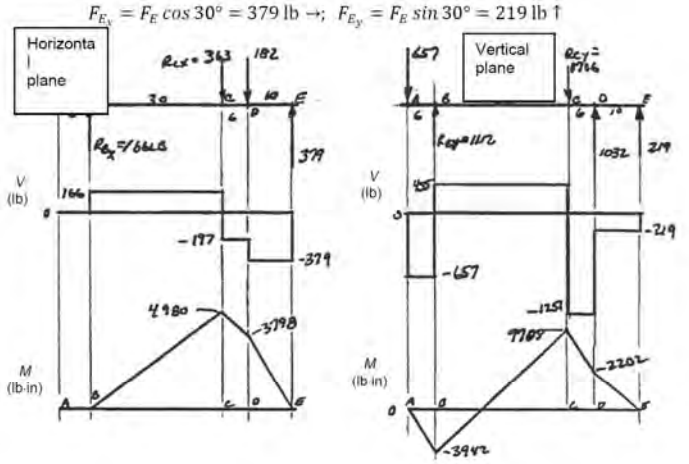
$$F_{D_x} = 985 \sin 30° \leftarrow -358 \cos 30° \rightarrow = 182 \text{ lb} \leftarrow$$

$$F_{D_y} = 985 \cos 30° + 358 \sin 30° = 1032 \text{ lb} \uparrow$$

Forces on Sprocket E:

$$F_E = \frac{T_E}{R_E} = \frac{1313}{3} = 438 \text{ lb}$$

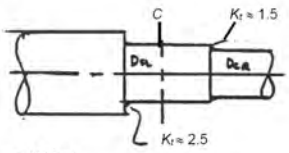
$$F_{E_x} = F_E \cos 30° = 379 \text{ lb} \rightarrow; \quad F_{E_y} = F_E \sin 30° = 219 \text{ lb} \uparrow$$



At C:

$T = 2625$ lb · in: Use $N = 4$ Shock

$M = \sqrt{4980^2 + 9708^2} = 10\,911$ lb · in



Choose 1137 CD; $s_y = 82\,000$ psi, $s_u = 98$ ksi, $s_n = 37$ ksi

Bearing seat

$s_n' = (0.8)(0.81)(37) = 24.0$ ksi

$$D_{CL} = \left(\frac{32(4)}{\pi} \sqrt{\left(\frac{2.4(10\,911)}{24\,000} \right)^2 + \frac{3}{4}\left(\frac{2625}{82\,000} \right)^2} \right)^{1/3} = 3.59 \text{ in}$$

Specify 3.75 in; $C_s = 0.76$ **Ok**

With $K_t = 1.5$; $D_{CR} = 3.03$ in; **Specify 3.20 in**; $C_s = 0.77$ **Ok**

Note: For the V-belt $F_{net} = F_1 - F_2 = 438$ lbs but total force is $F_1 + F_2 = 1.5F_{net} = 656$ lbs. (Mott Eqn. 12-12)
Note: Solution for position C above uses Safety Factor = 4. Spreadsheet below uses SF = 3.0 as stated in problem

	V-belt left	V-belt middle	V-belt right		Bearing B left	Bearing B middle	Bearing B right		Bearing C left	Bearing C middle	Bearing C right		Spur Gear left	Spur Gear middle	Spur Gear D right		Chain E left	Chain E middle	Chain E right	
Design Factor N	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Type of Stress Concentration	Ring	Profile Key	Ring		LG Fillet	None	Sharp Fillet		Sharp Fillet	Profile Key	LG Fillet		Ring	Profile Key	Ring		Ring	Profile Key	Ring	
Stress Concentration Kt	3	2	3		1.5	1	2.5		2.5	1	1.5		3	2	3		3	2	3	
Total Shear (lbs)	656	656	656		678	678	466		1307	1307	1307		1307	1307	1307		437	437	437	
Bending Moment M (lb-in)	0	0	0		3942	3942	3942		10910	10910	10910		4390	4390	4390		0	0	0	
Torque (lb.-in)	2625	2625	2625		2625	2625	2625		2625	2625	2625		3938	3938	3938		1313	1313	1313	
Corrected Endurance Limit S'n (psi	23976	23976	23976	23976	23976	23976	23976		23976	23976	23976		23976	23976	23976		23976	23976	23976	
Yield strenght (psi)	82000	82000	82000	82000	82000	82000	82000		82000	82000	82000		82000	82000	82000		82000	82000	82000	
Torque&Bending Diameter (in)12-24					1.96336	1.719811	2.324319		3.260112	2.40405	2.750636		2.560444	2.239702	2.560444		0.751321	0.751321	0.751321	
Alternating Shear Diameter (in)	0.850861	0.694725	0.850861																	
Adjusted Diameter (1.06D)	0.901912	0.736408	0.901912										2.714071		2.714071		0.7964		0.7964	
Diameter (mm,)	21.61186	17.64601	21.61186		49.86934	43.6832	59.0377		82.80684	61.06286	69.86615									
Selected Diamter (inches)	1.5	1.5	1.5	1.5				3.75				2.75	2.75	2.75	2.75	2.75	1.5	1.5	1.5	
Bearing Diameter (inches)					2.362205	2.362205	2.362205		3.346457	3.346457	3.346457									
					60	60	60		85	85	85									
					Bearing	6212	C10 Rating	10679	Bearing	6217	C10 Rating	18705								
					Min. Shoulder Dia. (mm,)			68	Min. Shoulder Dia. (mm,)			94								
					Max. Shoulder Dia. (mm)			102	Max. Shoulder Dia. (mm)			141								

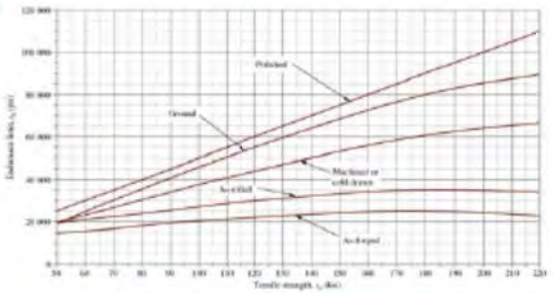
Design Equation for Shaft Design

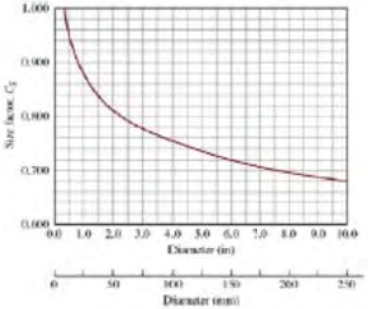
$$D = \left[\frac{32\,N}{\pi}\sqrt{\left[\frac{K_f M}{S'_n}\right]^2 + \frac{3}{4}\left[\frac{T}{S_y}\right]^2}\right]^{1/3} \quad (12\text{–}24)$$

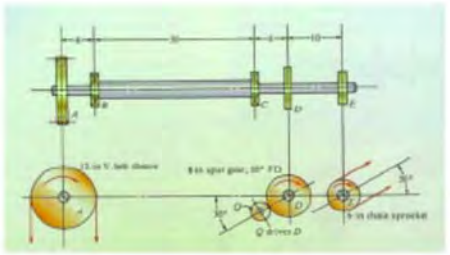
Required Shaft Diameter

$$D = \sqrt{2.94\,K_f V N / S_y} \quad (12\text{–}16)$$

Material	1020 CD	1137 CD	1050 CD	1137 OQT	4140 OQT1000
yield strength	51000	82000	84000	60000	152000
ultimate strenght	61000	98000	100000	87000	168000
Brinell Hardness	122	196	200	174	341
mod. machine strengt	22000	37000	37000	32000	59000
99% rel.	0.81	0.81	0.81	0.81	0.81
Size Factor	0.8	0.8	0.8	0.8	0.8
Corrected Endurance	14256	23976	23976	20736	38232







Keyway Analysis	Gear	Chain	V-Belts
Keyway size	0.75	0.25	0.25
Diameter	3	1	1.5
Torque	3938	1313	2625
Safety Factor	1.5	1.5	1.5
yield strength	54000	54000	54000
Length	0.194469	0.583556	0.777778

$$L = \frac{4TN}{DWS_y} = \frac{4(900\ \text{lb}\cdot\text{in})(4)}{(0.875\ \text{in})(0.188\ \text{in})(54\,000\ \text{psi})} = 1.62\ \text{in}$$

TABLE 11–1 Key Size vs. Shaft Diameter

U.S. inch sizes				SI metric sizes			
Nominal shaft diameter		Key dimensions		Nominal shaft diameter		Key dimensions	
Over (in)	To including (in)	Width, W (in)	Height, H (in)	Over (mm)	To including (mm)	Width, W (mm)	Height, H (mm)
5/16	7/16	3/32	3/32	6	8	2	2
7/16	9/16	1/8	1/8	8	10	3	3
9/16	7/8	3/16	3/16	10	12	4	4
7/8	1 1/4	1/4	1/4	12	17	5	5
1 1/4	1 3/8	5/16	5/16	17	22	6	6
1 3/8	1 3/4	3/8	3/8	22	30	8	7
1 3/4	2 1/4	1/2	1/2	30	38	10	8
2 1/4	2 3/4	5/8	5/8	38	44	12	8
2 3/4	3 1/4	3/4	3/4	44	50	14	9
3 1/4	3 3/4	7/8	7/8	50	58	16	10
3 3/4	4 1/2	1	1	58	65	18	11
4 1/2	5 1/2	1 1/4	1 1/4	65	75	20	12
5 1/2	6 1/2	1 1/2	1 1/2	75	85	22	14
6 1/2	7 1/2	1 3/4	1 1/2	85	95	25	14
7 1/2	9	2	1 1/2	95	110	28	16
9	11	2 1/2	2	110	130	32	18
11	13	3	2 1/2	130	150	36	20
13	15	3 1/2	3	150	170	40	22
15	18	4	3 1/2	170	200	45	25
18	22	5	4	200	230	50	28
22	26	6	4 1/2	230	260	56	32
26	30	7	5	260	290	63	32
30	38	8	6	290	330	70	36
				330	380	80	40
				380	440	90	45
				440	500	100	50

TABLE 11–4 Selection of Materials Used for Keys

Material designation	Tensile strength S_u (ksi)	Tensile strength S_u (MPa)	Yield strength S_y (ksi)	Yield strength S_y (MPa)
Carbon steels (SAE)				
1020	64	441	54	372
1040	91	626	59.5	372
1045	91	627	77	531
1095	140	965	83	572
Alloy steels (SAE)				
4140	150	703	90	620
4640	130	892	95	655
Stainless steels (SAE)				
303	90	621	35	240
304	85	586	35	241
316	85	586	35	241
416	75	517	40	275
Aluminum				
6061	18	124	12	83

Insert shear, bending moment and torque values into spreadsheet. See video demo on shaft sizing, bearing selection and key sizing for detailed information.

Notes:

1) Modified endurance strength for SAE 1137 CD steel having S_y = 82 ksi and S_u = 98 ksi must be determined. The machined value is S'_e = 37 ksi (Mott Fig. 5-11 @ S_u = 98ksi) with size factor estimated as 0.8 (Mott Fig. 5-12) and 99% reliability as 0.81 giving modified endurance strength as 37000 (0.8)(0.81) = 23900 psi.

2) The torque at the center of the spur gear is the full torque of 3938 lb-in.

3) The maximum diameter for the spur gear is determined as 2.72 inches (Eq. 12-24 with the 1.06D offset). This value is rounded up to 2.75 inches (Mott Table A2-1).
Therefore the solution to **Part c) is $D_{spur\ gear}$ = 2.750 inches.**

4) The torque at the V-belt is 10 HP which translates to 2625 lb.-in. The maximum diameter for the spur gear is determined as 0.902 inches (Eq. 12-24 with the 1.06D offset). This value would normally be rounded up to 1.00 inches (Mott Table A2-1). However, a diameter of 1.50 inches is chosen as the next step is quite large
Therefore the solution to **Part c) is $D_{V\text{-}belt}$ = 1.500 inches. (any reasonable value from 1.000 is acceptable)**

5) The torque at the Chain Drive is 5 HP which translates to 1313 lb-in. The maximum diameter for the chain drive is determined as 0.796 inches (Eq. 12-24 with the 1.06D offset). This value would normally be rounded up to 1.00 inches (Mott Table A2-1). However, a diameter of 1.50 inches is chosen as the next step is quite large
Therefore the solution to **Part c) is D_{chain} = 1.500 inches. (any reasonable value from 1.000 is acceptable)**

6) The sizing for keys is shown on the spreadsheet. The width area all selected based on the diameters determined in Part c). The length calculations based on the key material (SAE 1018 carbon steel) and safety factor of 4.0. The minimum key lengths for the spur gear is 0.679", the V-belt sheave is 1.38", and the chain sprocket is 0.691 inches. These values are normally increased to the axial length of the components. This information is not given. We can make some reasonable assumptions here. .
Part d) Choose Keyways i) V-Belt W = H = 0.375 inches by 1.5 inches. ii) Spur Gear W = H = 0.625 inches by 1.0 inches. iii) Chain Drive W = H = 0.375 inches by 1.0 inches.

7) Since there is torque and bending moment at the bearing locations, use the alternating bending moment and constant shear equation (Eq. 12-24) to determine the minimum diameters (2.324" for Bearing B (sharp fillet) and 3.260 for Bearing C (sharp fillet). Convert these values to millimeter for bearing selection (59 and 83 mm.). The minimum bearing bore that exceeds these values are 60 and 85 mm.

8) For bearing selection, first determine the expected lifetime for the bearing (see below) and use this value to determine the C_{10} values for the loads on the bearings (3500 lbs for the Bearing B and 6753 lbs for Bearing C). From Table 14-3, at a bore = 60 mm., we can select the 6012 bearing with a rated C_{10} = 6655 lbs. The minimum shaft shoulder diameter is 67 mm. and maximum is 88 mm. For Bearing C at a bore of 85 mm. we can select the 6017 bearing with a rated C_{10} = 11106 lbs. The minimum shaft shoulder diameter is 92mm. and maximum shoulder diameter is 123 mm.
However, the minimum shoulder diameter for Bearing C is larger than the maximum shoulder diameter for Bearing B. We want to have the diameter between B and C held constant. Hence, lets select the 6212 Bearing for Bearing B (C_{10} = 10679 lbs, min. dia. = 68mm. and max. dia. = 102mm). If we select D = 3.75 inches (95.25mm.) for between the bearings we can meet the minimum shoulder diameter at C.
Part e) Bearing B: Select 6212 Deep Groove Bearing (Bore = 60mm.) with min. shoulder dia. = 68 mm.
Bearing C: Select 6017 Deep Groove Bearing (Bore = 85mm.) with min. shoulder dia. = 92 mm.

9) Include the min. shaft diameters for the bearing seats for the prescribed fits in Table 15-5 (see below). The min. seat diameter of 2.3616" (bearing B) and 3.3457" (Bearing C) will meet the ISO tolerance fit for a k5 interference fit. Normally the diameter of the shaft prior to the bearing seats must be less than the seat diameter in order to position the bearings to the seats prior to press fitting.
L = 60 min/hr* 8 hr/day* 20 days/month * 60 months * 240 rpm = 1.4 (10^8) cycles

Bearing B Shear load = radial load = 678 lbs.
C_{10} = 678 (1.4 (10^8)/10^6)^0.333 = 3500 lbs
Bearing C Shear load = radial load = 1307 lbs.
C_{10} = 1307 (1.4 (10^8)/10^6)^0.333 = 6753 lbs

TABLE 15–5 Shaft and Housing Fits for Bearings

A. Shaft fits

Bearing bore			ISO tolerance grade	Shaft diameter		Limits of fit	
Nominal (mm)	Maximum (in)	Minimum (in)		Maximum (in)	Minimum (in)	Minimum (in)	Maximum (in)
10	0.3937	0.3934	j5	0.3939	0.3936	0.0001L	0.0005T
12	0.4724	0.4721	j5	0.4726	0.4723	0.0001L	0.0005T
15	0.5906	0.5903	j5	0.5908	0.5905	0.0001L	0.0005T
17	0.6693	0.6690	j5	0.6695	0.6692	0.0001L	0.0005T
20	0.7874	0.7870	k5	0.7878	0.7875	0.0001T	0.0008T
25	0.9843	0.9839	k5	0.9847	0.9844	0.0001T	0.0008T
30	1.1811	1.1807	k5	1.1815	1.1812	0.0001T	0.0008T
35	1.3780	1.3775	k5	1.3785	1.3781	0.0001T	0.0010T
40	1.5748	1.5743	k5	1.5753	1.5749	0.0001T	0.0010T
45	1.7717	1.7712	k5	1.7722	1.7718	0.0001T	0.0010T
50	1.9685	1.9680	k5	1.9690	1.9686	0.0001T	0.0010T
55	2.1654	2.1648	k5	2.1660	2.1655	0.0001T	0.0012T
60	2.3622	2.3616	k5	2.3628	2.3623	0.0001T	0.0012T
65	2.5591	2.5585	k5	2.5597	2.5592	0.0001T	0.0012T
70	2.7559	2.7553	k5	2.7565	2.7560	0.0001T	0.0012T
75	2.9528	2.9522	k5	2.9534	2.9529	0.0001T	0.0012T
80	3.1496	3.1490	k5	3.1502	3.1497	0.0001T	0.0012T
85	3.3465	3.3457	k5	3.3472	3.3466	0.0001T	0.0015T
90	3.5433	3.5425	k5	3.5440	3.5434	0.0001T	0.0015T
95	3.7402	3.7394	k5	3.7409	3.7403	0.0001T	0.0015T
100	3.9370	3.9362	k5	3.9377	3.9371	0.0001T	0.0015T
105	4.1339	4.1331	m5	4.1350	4.1344	0.0005T	0.0019T
110	4.3307	4.3299	m5	4.3318	4.3312	0.0005T	0.0019T
115	4.5276	4.5268	m5	4.5287	4.5281	0.0005T	0.0019T
120	4.7244	4.7236	m5	4.7255	4.7249	0.0005T	0.0019T
125	4.9213	4.9203	m5	4.9226	4.9219	0.0006T	0.0023T
130	5.1181	5.1171	m5	5.1194	5.1187	0.0006T	0.0023T
140	5.5118	5.5108	m5	5.5131	5.5124	0.0006T	0.0023T
150	5.9055	5.9045	m6	5.9071	5.9061	0.0006T	0.0026T
160	6.2992	6.2982	m6	6.3008	6.2998	0.0006T	0.0026T
170	6.6929	6.6919	m6	6.6945	6.6935	0.0006T	0.0026T
180	7.0866	7.0856	m6	7.0882	7.0872	0.0006T	0.0026T
190	7.4803	7.4791	m6	7.4821	7.4810	0.0007T	0.0030T
200	7.8740	7.8728	m6	7.8758	7.8747	0.0007T	0.0030T